

Enhancing Robustness in Federated Learning against Adversarial Attacks: A Loss-Based Client Clustering Approach

While Federated Learning (FL) aims to enhance privacy by preventing direct data exchange, it still faces **privacy** and **security** issues. In terms of **security**, FL is particularly susceptible to adversarial threats. These vulnerabilities often emerge early in the system's design. One key issue stems from the server's ability to access individual user updates, which, if exploited by a malicious server, can lead to manipulation of the training process and unauthorized control over global model parameters, ultimately compromising the model's integrity. Another risk arises from the dynamic nature of user participation—since users can enter or exit the system at will, adversaries may take advantage of this openness to observe global model updates, extract sensitive information, or intentionally disrupt the model's performance. From a security standpoint, attacks in FL primarily target **integrity** and **availability**. These types of threats aim to manipulate the model's output, potentially leading to significant disruptions in its intended functionality. **Integrity-based** attacks involve manipulating the model to consistently produce attacker-specified outputs during inference. On the other hand, threats to **availability** hinder users from accessing accurate model predictions or interfere with certain features, rendering the model unreliable for practical use [1]. When it comes to **privacy**, adversaries typically aim to compromise confidentiality. Attackers may seek to uncover sensitive aspects of the target model, including its internal parameters or prediction outputs [2]. Despite ongoing efforts to develop privacy-preserving mechanisms, studies have shown that user privacy remains vulnerable. For instance, during the training phase, a malicious client can infer specific training data points or extract attribute-level details from other participants' data, leading to the exposure of private information [3].

Among the various adversarial threats in FL, **poisoning attacks**—where adversarial participants deliberately inject malicious updates or manipulate local training data—pose a significant challenge by hindering convergence and undermining the integrity of the global model. Poisoning attacks in FL are typically categorized as either **targeted** or **untargeted**. Targeted attacks are aimed at disrupting or degrading the performance of the training process or the models of specific participants. A well-known example is **label-flipping**, where malicious clients alter their local training data by mislabeling class labels prior to local model training. The objective is to compromise the integrity of the global model by injecting structured label inconsistencies that lead to degraded performance. In contrast, untargeted attacks affect the entire system without focusing on particular participants, models, or tasks. Common examples include **sign-flipping**, where Byzantine clients adversarially alter their local model updates prior to transmission to the server. In particular, they invert the sign of each element in the gradient vector, thereby reversing the direction of the update and impairing the convergence of the global model. Another example is **Gaussian noise injection** where Byzantine clients perturb their local model updates by injecting random noise prior to the server-side aggregation.

To address the threat of adversarial attacks within FL, it is essential to adopt **robust aggregation methods** designed to be resilient to malicious behavior. Robust aggregators incorporate strategies to

detect and exclude malicious inputs from malicious participants. These methods typically rely on techniques such as identifying outliers, applying robust statistical measures, and setting thresholds to filter out harmful updates, thus maintaining the integrity of the global model even when adversaries are present. Several aggregation operators based on robust estimators have been proposed, including the following: a) **Mean** [4]: Serves as the standard aggregation technique in FL, where the central server computes the element-wise average of the model updates received from all participating clients. However, it lacks robustness against adversarial inputs: a single malicious client can significantly distort the average due to its sensitivity to outliers and lack of robustness guarantees. This vulnerability renders Mean aggregation susceptible to various poisoning attacks, including sign flipping, label flipping, and noise injection strategies. As such, it is often used as a baseline for benchmarking more robust aggregation rules designed to mitigate adversarial influence; b) **Trimmed Mean** [5]: Serves as an alternative to standard mean aggregation, employing a coordinate-wise aggregation rule; c) **Median** [5]: Another coordinate-wise aggregation rule, following the steps of Trimmed Mean. For each model parameter (i.e., each dimension of the gradient vector), it computes the median of the corresponding values received from all clients; d) **Krum** [6]: Krum operates by evaluating each client's model update based on its consistency with those of other clients; and e) **Multi-Krum** [6]: Multi-Krum extends the Krum aggregation rule by selecting multiple client updates rather than a single one, thereby improving convergence stability while maintaining Byzantine robustness. While the aforementioned defense schemes represent many of the current state-of-the-art approaches, the research community has increasingly shifted focus toward uncovering additional factors that influence robust aggregation in FL.

To address the challenge of adversarial behavior in decentralized FL, while also aligning with **CoEvolution's objectives** to develop protection mechanisms that enhance the security of training pipelines—particularly against adversarial examples and poisoning attacks—**ICCS proposes a robust aggregation mechanism** which leverages **client clustering** based on model performance (loss values) to distinguish between benign and potentially malicious updates. By comparing **client-submitted models against the server's trusted baseline**, poisoned contributions can be effectively isolated, thereby promoting **convergence**. The approach assumes at least two honest clients, homogeneously distributed data, and full client participation in each round. Despite its simplicity and minimal trust requirements, the method demonstrates strong empirical robustness and accuracy across various attack types and datasets, outperforming state-of-the-art baselines. More specifically, experimental evaluations across CIFAR-10, FMNIST, and MNIST datasets demonstrates that the proposed defense consistently **outperforms** classical aggregation strategies and robust baselines under various adversarial attack scenarios—including label-flipping, sign-flipping, and Gaussian noise injection. Unlike existing methods, which either aggregate all client updates or rely on simplistic statistical or distance-based criteria, ICCS's approach dynamically **filters out malicious contributions using client-side loss clustering**. This results in **higher centralized accuracy** and **stable convergence**, even when half of the participating clients are adversarial. Importantly, the method does not require prior knowledge of the number of attackers and remains robust across diverse attack types, validating its practical effectiveness and generalization potential in decentralized FL settings.

References

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